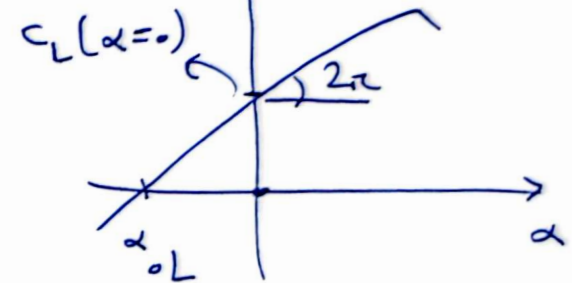
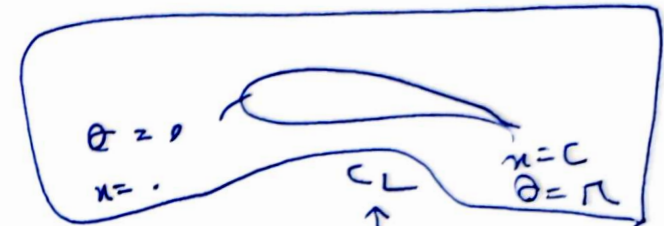
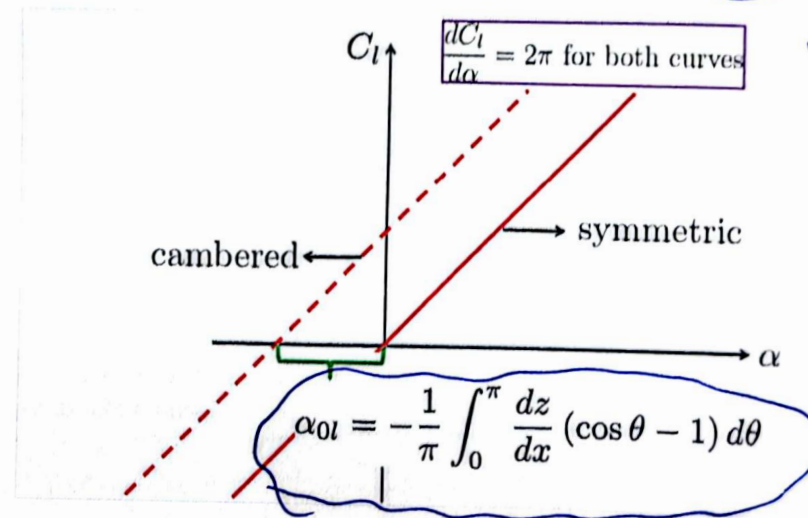


06-02-2025

Thin airfoil theory summary
 " " " examples
 3D wings

Thin airfoil theory - coefficients



$\frac{dz}{dx}$ = "slop of camber"

$$C_l = 2\pi (\alpha - \alpha_{0L})$$

Thin airfoil theory - derived quantities

Moment coefficient:

$$m_0 = - \int_0^c \xi dl = -\rho U_\infty \int_0^c \xi \gamma(\xi) d\xi \quad (4.63)$$

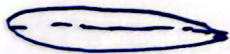
$$C_{m0} = - \left[\frac{C_l}{4} + \frac{\pi}{4} (A_1 - A_2) \right] \quad (4.64)$$

$$= -\frac{\pi}{2} \left(A_0 + A_1 - \frac{A_2}{2} \right) \quad (4.65)$$

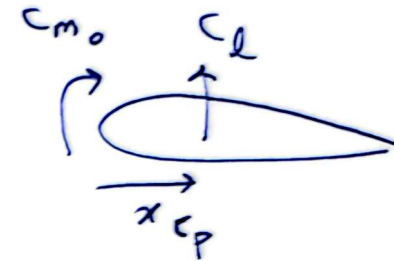
Symmetric airfoil, $A_0 = \alpha$, $A_1 = A_2 = 0$. Therefore in the case of a symmetric airfoil

for sym. airfoil $\rightarrow$

$$C_{m0} = -\frac{\pi}{2} (\alpha) = -\frac{2\pi\alpha}{4} = -\frac{C_l}{4} \quad (4.66)$$



A_0
 A_1
 A_2



Thin airfoil theory - derived quantities

Center of pressure:

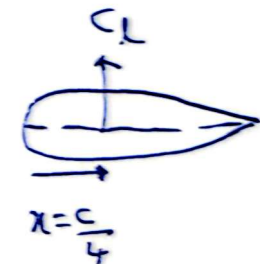
$$x_{cp} = c \frac{-C_{m0}}{C_l} \quad (4.67)$$

$$\frac{x_{cp}}{c} = \frac{1}{4} \left[1 + \frac{\pi}{C_l} (A_1 - A_2) \right] \quad (\text{TEq. (6.27)}) \quad (4.68)$$

Note: Here the location of x_{cp} depends on C_l which in turn is a function of α . Therefore the location of the center of pressure changes with the angle of attack α . However, in the case of the symmetric airfoil,

$$\rightarrow \text{Symmetric a/f} \rightarrow C_{m0} = -\frac{C_l}{4} \rightarrow x_{cp} = \frac{c}{4} \quad (4.69)$$

Therefore, in the case of a symmetric airfoil, the center of pressure is independent of the angle of attack



Thin airfoil theory - derived quantities

Aerodynamic center: Use transfer for moments i.e., eq. (4.27)

$$m_x = xl + m_0$$

$$C_{mx} = \frac{x}{c} C_l + C_{m0} \quad (4.70)$$

$$\text{choose } x = \frac{c}{4} \rightarrow C_{mc/4} = \frac{C_l}{4} + C_{m0} \quad (4.71)$$

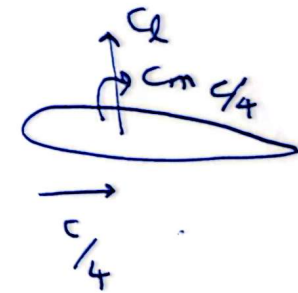
$$C_{mc/4} = \frac{\pi}{4} (A_2 - A_1) \quad (\text{TEq. (6.29)}) \quad (4.72)$$

$$C_{mc/4} = \frac{\pi}{4} (A_2 - A_1), \text{ depends on camber but not } \alpha \quad (4.73)$$

$$\text{Therefore, } \underline{x_{ac} = \frac{c}{4}} \text{ \& } C_{mac} = C_{mc/4} \quad (4.74)$$

Note, for a symmetric airfoil

$$C_{mac} = C_{mc/4} = C_{mx_{cp}} = 0 \quad (4.75)$$



$c/4$ is x_{a-c}

For sym. airfoil, ~~can~~ $\frac{c}{4}$ is x_{cp} $\rightarrow$ $x_{a.c}$

Thin airfoil theory - Summary

$$z(x) \neq 0$$

$$\gamma(\theta) = 2U_\infty \left[A_0 \frac{(1 + \cos \theta)}{\sin \theta} + \sum_{n=1}^{\infty} A_n \sin(n\theta) \right]$$

$$C_l = 2\pi(\alpha - \alpha_{ZL}) = 2\pi \left(A_0 + \frac{A_1}{2} \right), \quad \frac{dC_l}{d\alpha} = 2\pi$$

$$C_{m0} = - \left[\frac{C_l}{4} + \frac{\pi}{4} (A_1 - A_2) \right]$$

$$\frac{x_{cp}}{c} = \frac{1}{4} \left[1 + \frac{\pi}{C_l} (A_1 - A_2) \right]$$

$$\frac{x_{ac}}{c} = \frac{1}{4}$$

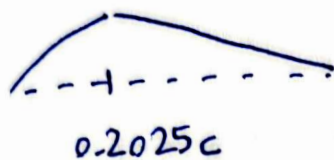
$$C_{mac} = C_{mc/4} = \frac{\pi}{4} (A_2 - A_1)$$

$$C_d = 0$$

$$\text{Symmetric airfoil} \rightarrow A_0 = \alpha, \quad A_{n \geq 1} = 0$$

Ex.1) An airfoil with camber profile

$$\begin{cases} \frac{z}{c} = 2.6595 \left[\left(\frac{x}{c}\right)^3 - 0.6075 \left(\frac{x}{c}\right)^2 + 0.11471 \left(\frac{x}{c}\right) \right] & \textcircled{1} \\ & 0 < x < 0.2025c \\ \frac{z}{c} = 0.022083 \left(1 - \frac{x}{c}\right) & \textcircled{2} \quad 0.2025c < x < c \end{cases}$$



a) Calculate A_0, A_1, A_2 , b) calculate c_L at $\alpha = 4^\circ$

c) " α_{L_0}

Solution: 1) Compute $\frac{dz}{dx}$ as given

$$\frac{dz}{dx} = \begin{cases} 2.6595 \left[3 \left(\frac{x}{c}\right)^2 - 2 \times 0.6075 \left(\frac{x}{c}\right) + 0.11471 \right] & 0 < \frac{x}{c} < 0.2025 \\ -0.022083 & 0.2025 < \frac{x}{c} < 1 \end{cases}$$

2) bring $\left\{ \frac{dz}{dx} \right\}$ to polar, and limits, $x = \frac{c}{2} (1 - \cos \theta)$ (known)

$$\boxed{\frac{x}{c} = \frac{1}{2} - \frac{1}{2} \cos \theta}$$

①

• $0 < \frac{x}{c} \leq 0.2025$ (first part of camber)

$$0 < \frac{1}{2} - \frac{1}{2} \cos \theta \leq 0.2025$$

$$\frac{-1}{2} < \frac{-1}{2} \cos \theta \leq -0.2975 \quad \div \frac{-1}{2}$$

$$1 < \cos \theta < 0.595 \implies$$

$$0 < \theta \leq 0.9335 \text{ rad}$$

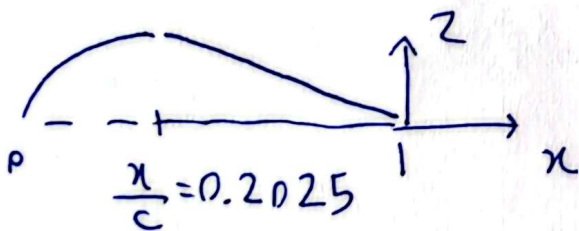
We need θ in radians
not degree

Second part of camber

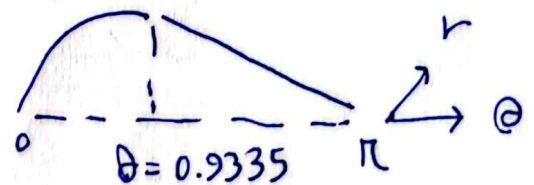
$$0.2025 < \frac{x}{c} < 1$$

Again replace $\frac{x}{c} = \frac{1}{2} - \frac{1}{2} \cos \theta \longrightarrow$ Solve

$$0.9335 \leq \theta \leq \pi$$



Cartesian



Polar

(2)

2) Convert $\frac{dz}{dx}$ to polar, put $\frac{x}{c} = \frac{1}{2} - \frac{1}{2} \cos \theta$ in
 ① and ② (do so, simplify)

$$\Rightarrow \frac{dz}{dx} = \begin{cases} 0.6840 - 2.3736 \cos \theta + 1.995 \cos^2 \theta & \cdot \theta < 0.9335 \\ -0.02208 & 0.9335 < \theta < \pi \end{cases}$$

③ 3) compute integrals:

$$A_1 = \frac{2}{\pi} \int_0^{\pi} \frac{dz}{dx} \cos \theta = \frac{2}{\pi} \int_0^{0.9335} (0.6840 - 2.3736 \cos \theta + 1.995 \cos^2 \theta) \cos \theta d\theta + \frac{2}{\pi} \int_{0.9335}^{\pi} (-0.02208) \cos \theta d\theta$$

$\frac{dz}{dx}$
 $\frac{dz}{dx}$

$$= \frac{2}{\pi} [0.1324 + 0.0177] = 0.0956$$

$$A_2 = \frac{2}{\pi} \int_0^{\pi} \frac{dz}{dx} \cos^2 \theta = \frac{2}{\pi} \int_0^{0.9335} \left(\frac{dz}{dx} \right) \cos^2 \theta + \int_{0.9335}^{\pi} \left(\frac{dz}{dx} \right) \cos^2 \theta = 0.0792$$

$\frac{dz}{dx}$
 $\frac{dz}{dx}$

(P-3)

$$\alpha_{L_1} = -\frac{1}{\pi} \int \frac{dz}{dx} (\cos\theta - 1) = -\frac{1}{\pi} \int_{0.9335}^{0.9335} (-0.6840 + 3.0576 \cos\theta - 4.3686 \cos^2\theta + 1.995 \cos^3\theta) d\theta - \frac{1}{\pi} \int_{0.9335}^{\pi}$$

$$(0.02208 - 0.02208 \cos\theta) d\theta = \underbrace{-0.0191 \text{ rad}}_{\substack{\text{the result is} \\ \text{in radian}}} = \underbrace{-1.09^\circ}_{\substack{\text{convert} \\ \text{to degree}}}$$

radiant

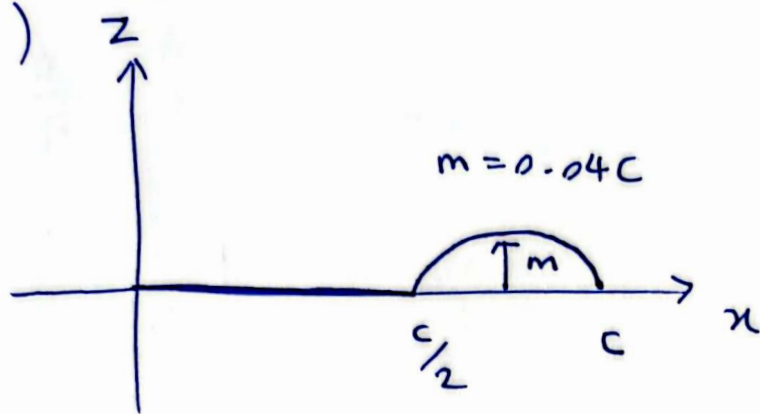
$$C_L = 2\pi (\alpha - \alpha_{L_1}) = 2\pi (\underbrace{0.0698}_{=4^\circ} + 0.0191) = 0.56$$

$$A_0 = \alpha - \frac{1}{\pi} \int_{0.9335}^{\pi} \frac{dz}{dx} d\theta = 0.0698 - \frac{1}{\pi} \int_{0.9335}^{0.9335} 0.6840 - 2.3736 \cos\theta + 1.995 \cos^2\theta - \frac{1}{\pi} \int_{0.9335}^{\pi} -0.02208 d\theta$$

$$= 0.04$$

Note: $\cos^2\theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$

Ex. 2)



$$z(x) = \begin{cases} 0 & (0 \leq x \leq \frac{c}{2}) \\ \frac{16m}{c^2} (x - \frac{c}{2})x & (\frac{c}{2} \leq x \leq c) \end{cases}$$

Find: A_1, A_2, α_L, c_L at $\alpha = 3^\circ$, $c_m \%$

Solution: 1) Convert compute $\frac{dz}{dx}$

$$z = \begin{cases} 0 & 0 \leq x \leq \frac{c}{2} \\ \frac{24mx}{c} - 16m\frac{x^2}{c^2} - 8m & \frac{c}{2} \leq x \leq c \end{cases}$$

$$\frac{dz}{dx} = \begin{cases} 0 & 0 \leq x \leq \frac{c}{2} \\ \frac{24m}{c} - 32m\frac{x}{c^2} & \frac{c}{2} \leq x \leq c \end{cases}$$

↑
Cartesian

$$\frac{dz}{dx} = \begin{cases} 0 & 0 \leq x \leq \frac{\pi}{2} \\ \frac{24m}{c} - 32\frac{m}{c} \left(\frac{1}{2} - \frac{1}{2} \cos \theta \right) & \frac{\pi}{2} \leq x \leq \pi \end{cases}$$

Polar

(P.5)

$$A_1 = \frac{2}{\pi} \int_0^{\pi} \frac{dz}{dx} \cos \theta d\theta = \frac{2}{\pi} \left[\int_0^{\pi/2} \cancel{\cos \theta d\theta} + \int_{\pi/2}^{\pi} \left(24 \frac{m}{c} - 32 \frac{m}{c} \left(\frac{1}{2} - \frac{1}{2} \cos \theta \right) \right) \cos \theta d\theta \right] = \frac{2}{\pi} \int_{\pi/2}^{\pi} \left(24 \frac{m}{c} \cos \theta - \frac{16m}{c} \cos \theta + \frac{16m}{c} \cos^2 \theta \right) d\theta$$

$$= \frac{2}{\pi} \int_{\pi/2}^{\pi} \left(\frac{8m}{c} \cos \theta + \frac{16m}{c} \cos^2 \theta \right) d\theta = \frac{2}{\pi} \left[-\frac{8m}{c} + \frac{16m}{c} \times \frac{\pi}{4} \right] = \frac{16m}{\pi c} \left(\frac{\pi}{2} - 1 \right) = A_1$$

$\frac{dz}{dx}$ in polar

$$A_2 = \frac{2}{\pi} \int_0^{\pi} \frac{dz}{dx} \cos(2\theta) d\theta = \frac{-32m}{3\pi c}$$

$$\alpha_{L_0} = -\frac{1}{\pi} \int_0^{\pi} \frac{dz}{dx} (\cos \theta - 1) d\theta = -\frac{1}{\pi} \int_{\pi/2}^{\pi} \left(\frac{8m}{c} \cos \theta - \frac{8m}{c} + \frac{16m}{c} \cos^2 \theta - \frac{16m}{c} \cos \theta \right) d\theta = -0.1 \text{ rad}$$

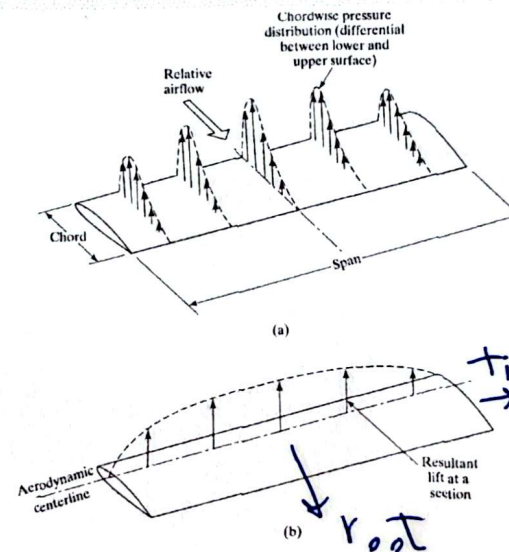
$$C_L = 2\pi (\alpha - \alpha_{L_0}) = 2\pi (0.05 - (-0.1)) = 0.95$$

rad

$$C_{m_{a.c}} = C_{m_{C/4}} = \frac{\pi}{4} (A_2 - A_1) = \frac{\pi}{4} \left[\frac{-32m}{3\pi c} - \frac{8m}{c} + \frac{16m}{\pi c} \right] = \frac{\pi}{4} \left[\frac{16m - 24\pi m}{3\pi c} \right], \quad m = 0.04c \Rightarrow C_{m_{a.c}} = -0.027$$

Potential Flow over Wings

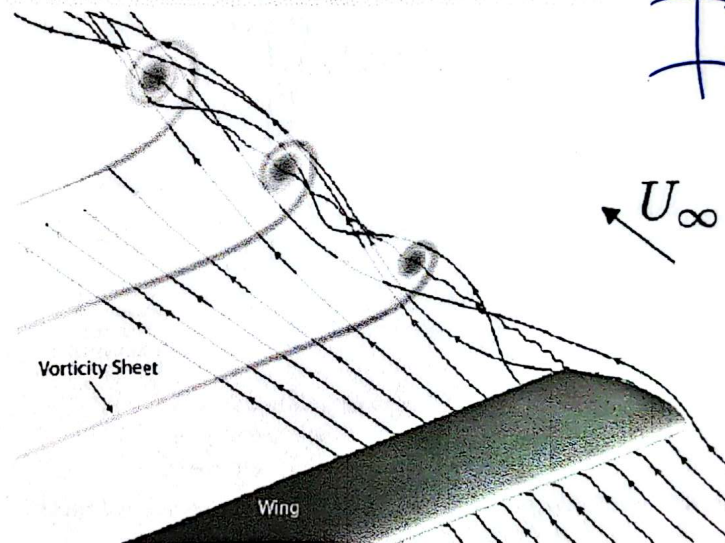
Lift distribution



lift is zero at tip
lift is Max at root

Figure 5.1: Flow over wings - qualitative effects, Fig. 7.1 from AE-BC

Wingtip vortices



<https://upload.wikimedia.org/wikipedia/commons/1/1c/Tip.vortex.rollup.png>

AE 319, EPG Pot. Wings, Qual. Effects, 201

Navigation icons: back, forward, search, and other presentation controls.

Downwash & Induced Drag

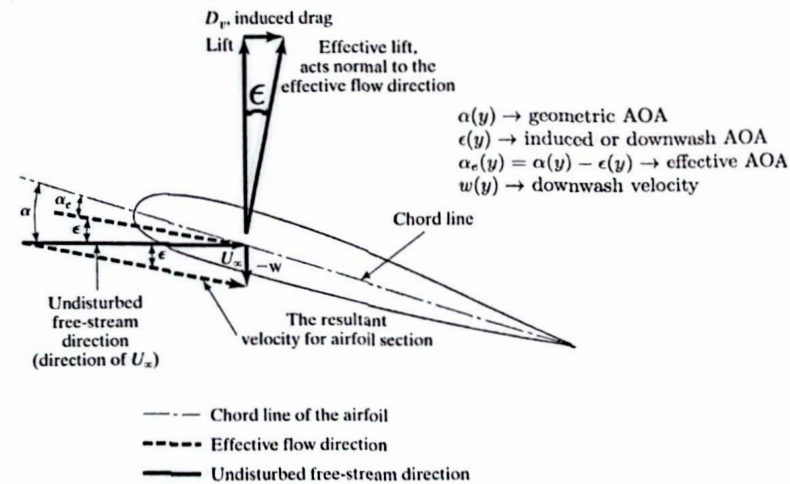
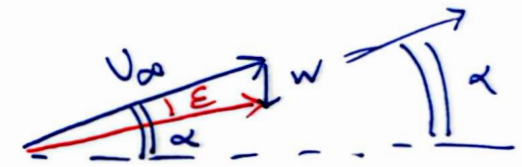
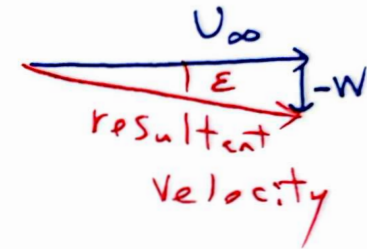
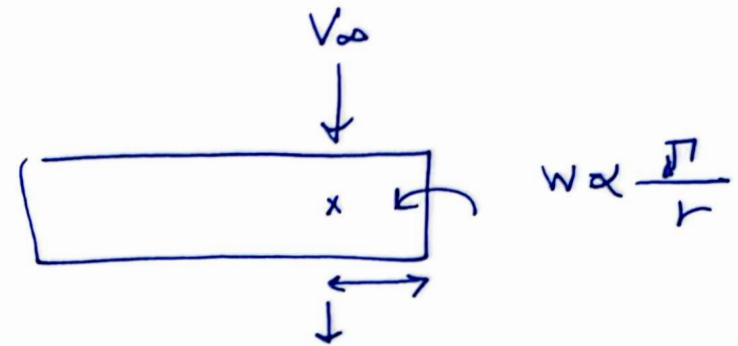


Figure 5.4: Downwash velocity and its effects, modified Fig. 7.6 from AE-BC

AE 319, EPG Pot. Wings, Qual. Effects, 202



$$\alpha_{eff} = \alpha - \epsilon$$

Airfoil experience
 α_{eff} due to downwash!

w changes with location

$w \propto \frac{\Gamma}{r}$
 $r = \dots$ @ tip $\rightarrow w$ is high at tip
 $r = \frac{b}{2}$ @ root " " small " roots